Khuat Thi Thu Hang

Robotics, Unmanned Aerial Vehicles, Path Planning

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Education

Vietnam National University — **Hanoi, Vietnam**
Master of Science in Electronics Engineering — *Feb. 2024 – Sep. 2024*

- CGPA: 3.62/4.0
- Thesis title: Distributed Motion Planning and Target Tracking in Multi-UAV Systems.

Vietnam National University — **Hanoi, Vietnam**
Bachelor of Engineering in Robotics Engineering — *Aug. 2019 – Jun. 2023*

- CGPA: 3.84/4.0 (Top 1%).
- Thesis title: Polar Coordinate-based Differential Evolution for Moving Target Search Using Vision Sensor on Unmanned Aerial Vehicles

Programming and other Hands-on Experience

- C/C++, Python, Matlab programming especially for robotics, UAV system, planning algorithm and swarm optimization.
- Programming and fusion Robot Operating System (ROS1 & ROS2).
- Using Gazebo for simulating complex environments and testing UAV/robot systems.
- Experience in containerization using Docker.
- 3D design in Solidworks.
- Skilled in Git, Latex.

Work Experiences

UAV Center - Viettel High Tech — **Hanoi, Vietnam**
Automation Control Engineer at VHT, full-time — *Feb. 2026 – Aug.2026*

- **Automatic take-off and landing for U3-UAV**
  - Integrate Total Energy Control System into the U3-UAV control system (SIL).
  - Design abort landing phase in U3-UAV system (SIL).

PhenikaaX — **Hanoi, Vietnam**
Drone Embedded Software Engineer at PhenikaaX, full-time — *Jul. 2023 – Dec.2025*

- **Autonomous Forest Survey with VTOL-UAV**
  - Integrate PX4 Autopilot software for VTOL, and perform flight log analysis for optimization.
  - Develop software to control the gimbal's orientation.
  - Develop the precision landing algorithm using computer vision.
  - Collect image data for further analysis.
- **Drone in Box**
  - Design autonomous flight behaviors, including mission planning.
  - Implement real-time data transmission and telemetry between the drone and server.
  - Develop drone software allowing server remote control of drone flight and mission execution.
  - Develop precision landing algorithm using computer vision and RTK-GPS.

Vietnam National University **Hanoi, Vietnam**
Research Study Assistant at Faculty of Electronics and Telecommunications *Jul. 2020 – Present*

- **Project: Pick-and-Place Robot for Autonomous Testing of Camera Module used in Smartphones**
  - Design HMI interface and develop for the PLC industrial computer
  - Perform fault extraction on factory test component camera via UART protocol
- **Project: Multi-goal Rapidly Exploring Random Tree with Safety and Dynamic Constraints for UAV Cooperative Path Planning**
  - Improved UAV's dynamics constrained in the RRT path planning algorithm
  - Validation through real-world UAV flight experiments
- **Project: Developing the Differential Evolution algorithm, establishing path planning for UAVs, applying for Search and Rescue**
  - Modeling the search and rescue problem: Continuous search environment, using a search camera sensor
  - Develop Differential Evolution algorithm to create an optimal flight path in search and rescue
- **Project: A sensor fusion approach for improving the implementation speed and accuracy of RTAB-Map algorithm-based indoor 3D mapping**
  - Application of the RTAB-Map algorithm in 3D mapping for mobile robots
  - Fusion of localization data using camera and IPS to enhance the accuracy of 3D mapping

Honors and Awards

Best Paper Award in 7th International Conference on Control, Robotics and Informatics	2024
Master Scholarship of Vingroup Innovation Foundation (VINIF)	2023, 2024
Third Prize, Science and Technology Award for Students Member of the Ministry of Education	2022
Scholarship of Odon Vallet	2022

Publications

- Thu Hang Khuat, Duy-Nam Bui, Thuy Ngan Duong, Manh Duong Phung. **Polar coordinate-based differential evolution for moving target search using vision sensor on unmanned aerial vehicles**, In *Intelligent Service Robotics*, 2026 (IF: 5.9 - Q1).
- Thu Hang Khuat, Duy-Nam Bui, Hoa TT Nguyen, Mien L Trinh, Minh T Nguyen, Manh Duong Phung. **Multi-goal Rapidly Exploring Random Tree with Safety and Dynamic Constraints for UAV Cooperative Path Planning**, In *IEEE Transactions on Vehicular Technology*, 2025 (IF: 7.5 - Q1).
- Duy-Nam Bui, Thu Hang Khuat, Manh Duong Phung, Thuan Hoang Tran, Dong LT Tran. **Model Predictive Control for Optimal Motion Planning of Unmanned Aerial Vehicles**, In *2024 7th International Conference on Control, Robotics and Informatics (ICCRI)*, 2024.
- Phan, H. A., Van Tan Duong, M. N. T., Thi, A. N., Thu, H. K. T., Duc, T. L., Van Hieu Dang, H. Q. D., ... , Bui, T. T. **Development of an Autonomous Component Testing System with Reliability Improvement Using Computer Vision and Machine Learning**, In *ECTI Transactions on Computer and Information Technology (ECTI-CIT)*, 2024.
- Phan, H. A., Nguyen, P. V., Khuat, T. H. T., Van, H. D., Tran, D. H. Q., Dang, B. L., ... Duc, T. C. **A sensor fusion approach for improving implementation speed and accuracy of RTAB-Map algorithm based indoor 3D mapping**, In *20th IEEE International Joint Conference on Computer Science and Software Engineering (JCSSE)*, 2023.

Reviewer

- IEEE Transactions on Industrial Electronics Journal.
- IEEE Transactions on Vehicular Technology.

References

- Dr. Manh Duong Phung –Assistant Professor – Vin University, Hanoi, Vietnam. e-mail: duong.pm@vinuni.edu.vn
- Dr. Van Nguyen Thi Thanh – Associate Dean, Faculty of Electronics & Telecommunications – Vietnam National University. e-mail: vanntt@vnu.edu.vn